Universidad de Cantabria

Master's Degree Thesis

SEARCHING FOR μ-DISTORTION IMPRINTS IN THE THERMAL SZ EFFECT FROM GALAXY CLUSTERS

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Abstract

While the cosmic microwave background (CMB) radiation is an almost perfect blackbody, small deviations can arise in the early Universe (μ-distortions) from various physical processes and later via the thermal Sunyaev-Zeldovich (SZ) effect (y-distortions), where CMB photons scatter off hot electrons in galaxy clusters. Because the SZ effect involves scattered CMB radiation, any primordial μ-distortion would also imprint a corresponding μ-dependent modification in the SZ signal. The project will compute how μ-type distortions modify the thermal SZ effect and then apply component separation techniques, such as the Constrained Internal Linear Combination (CILC)—using a Taylor expansion around $\mu = 0$—to extract this signal from galaxy cluster data. Using sky simulations for future CMB surveys, the study will assess whether this SZ-based approach can improve current upper limits on μ.

Resumen

Aunque la radiación del fondo cósmico de microondas (CMB) es un cuerpo negro casi perfecto, pueden surgir pequeñas desviaciones en el Universo temprano (distorsiones tipo μ) debido a diversos procesos físicos, y más tarde a través del efecto Sunyaev-Zeldovich (SZ) térmico (distorsiones tipo y), donde los fotones del CMB se dispersan al interactuar con electrones calientes en cúmulos de galaxias. Dado que el efecto SZ implica radiación dispersada del CMB, cualquier distorsión primordial tipo μ también dejaría impresa una modificación correspondiente dependiente de μ en la señal SZ. El proyecto calculará cómo las distorsiones de tipo μ modifican el efecto SZ térmico para luego aplicar técnicas de separación de componentes, tales como la Combinación Lineal Interna Constringida (CILC) —utilizando un desarrollo de Taylor alrededor de $\mu = 0$— para extraer esta señal a partir de datos de cúmulos de galaxias. Utilizando simulaciones del cielo para futuros cartografiados del CMB, el estudio evaluará si este enfoque basado en el efecto SZ puede mejorar los límites superiores actuales para μ.

Contents

1. Introducció

Remember Moncho's advice of using setnewcommand or newcommand to define new commands and make the writing faster.

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1.1 Spectral distribution of the CMB

The CMB is the afterglow of the Big Bang, a relic radiation that fills the universe and provides a snapshot of the early cosmos. It is a nearly perfect blackbody spectrum, with a temperature of approximately 2.725 K. The spectral distribution of the CMB can be described by Planck's law, which gives the intensity of radiation as a function of frequency or wavelength.

The current upper limits on μ-distortions from COBE-FIRAS are of the order of $|\mu| < 9 \times 10^{-5}$, which means that any deviations from a perfect blackbody spectrum must be very small. However, the ΛCDM model predicts that μ-distortions should be present at a level of around 10^{-8}, which is well below the current observational limits. Detecting μ-distortions would provide valuable insights into the physical processes that occurred in the early universe, such as energy release from decaying particles, dissipation of acoustic waves, or other non-standard processes that could have injected energy into the photon-baryon plasma before recombination

1.2 SZ effect

The Sunyaev-Zeldovich (SZ) effect is a distortion of the CMB spectrum caused by the inverse Compton scattering of CMB photons off high-energy electrons in galaxy clusters. This effect can be used to study the properties of galaxy clusters and the large-scale structure of the universe. The thermal SZ effect, in particular, is sensitive to the pressure of the intracluster medium and can be used to measure the cluster's gas content and temperature. Since the SZ effect involves the scattering of CMB photons, any primordial μ-distortion would also imprint a corresponding μ-dependent modification in the SZ signal. This means that if we can detect the SZ effect from galaxy clusters, we can also look for the presence of μ-distortions in the SZ signal, which could provide a new way to constrain μ-distortions and gain insights into the early universe.

1.3 μ-distorted SZ effect

The μ-distorted SZ effect refers to the modification of the thermal SZ effect due to the presence of μ-type distortions in the CMB spectrum. If the CMB photons that are scattered by the hot electrons in galaxy clusters have a μ-distortion, then the resulting SZ signal will also be modified in a way that depends on the value of μ. This means that by analyzing the SZ signal from galaxy clusters, we can potentially detect or constrain the presence of μ-distortions in the CMB.

1.4 Other effects

There are many other effects that can modify the CMB spectrum, such as y-distortions from the thermal SZ effect, kinematic SZ effect, and foreground emissions from our galaxy and extragalactic sources **.** These effects can complicate the detection of μ-distortions, as they can mimic or mask the signal we are trying to detect. Therefore, it is important to use component separation techniques, such as the Constrained Internal Linear Combination (CILC), to isolate the μ-distorted SZ signal from these other contributions. By applying these techniques to sky simulations for future CMB surveys, we can assess whether this SZ-based approach can improve current upper limits on μ-distortions and potentially lead to a detection of these distortions in the future.

2. Simulations

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In this chapter, I will describe the simulations I have performed to generate the sky maps used in this thesis.

The data used in this thesis are based on the CMB-HD sky simulations, which include multiple frequency channels, almost realistic noise levels, and other effects, but no foregrounds. These simulations are designed to mimic the observations that will be made by future CMB experiments, allowing us to test our analysis methods and assess their performance in a realistic setting.

There are two main groups of datasets used in this thesis: the first one is the set of 3-layer maps, and the second one is the set of 4-layer maps. The latter consist of the CMB, the thermal SZ effect, the μ-distortions, and instrumental noise, while the 3-layer maps are the result of removing one of these components at a time.

This allowed me to test the performance of the Constrained Internal Linear Combination (CILC) method in isolating the μ-distorted SZ signal from the other components, and to assess the impact of each component on the final results.

This served as a proof of concept for the analysis methods used in this thesis, and provided insights into the challenges and limitations of detecting μ-distortions in the SZ signal from galaxy clusters. The Internal Linear Combination (ILC), as will be shown in next chapters, is not able to isolate the μ-distorted SZ signal from the other components as there is a leakage of the thermal SZ effect into the reconstructed μ-distorted SZ map, which motivates the use of the CILC method to achieve a cleaner separation of these components.

The data is first downloaded from **WebSky_Mocks**, which comes in a HEALPix format with a resolution of $N_{side} = 4096$ and a pixel size of 0.86 arcmin. However, for the seek of computational efficiency and because of the resources available for this thesis, which is basically my laptop, I downgraded the resolution to $N_{side} = 2048$ and a pixel size of 1.72 arcmin. Then, the data is processed with the frequency scaling functions described in Zegeye, Crawford, and Hu, 2024 to generate the tSZ and the μ-distorted SZ maps, which can be seen in Fig. x.

The resulting maps are then convolved with a Gaussian beam to simulate the instrumental response, and noise is added to mimic the sensitivity of future CMB experiments and the maps are saved as 'data_cube' .fits files. These are used as input for the component separation methods described in the (Methodology) section, and the results are analyzed in the (Results) section to assess the performance of these methods in isolating the μ-distorted SZ signal from the other components.

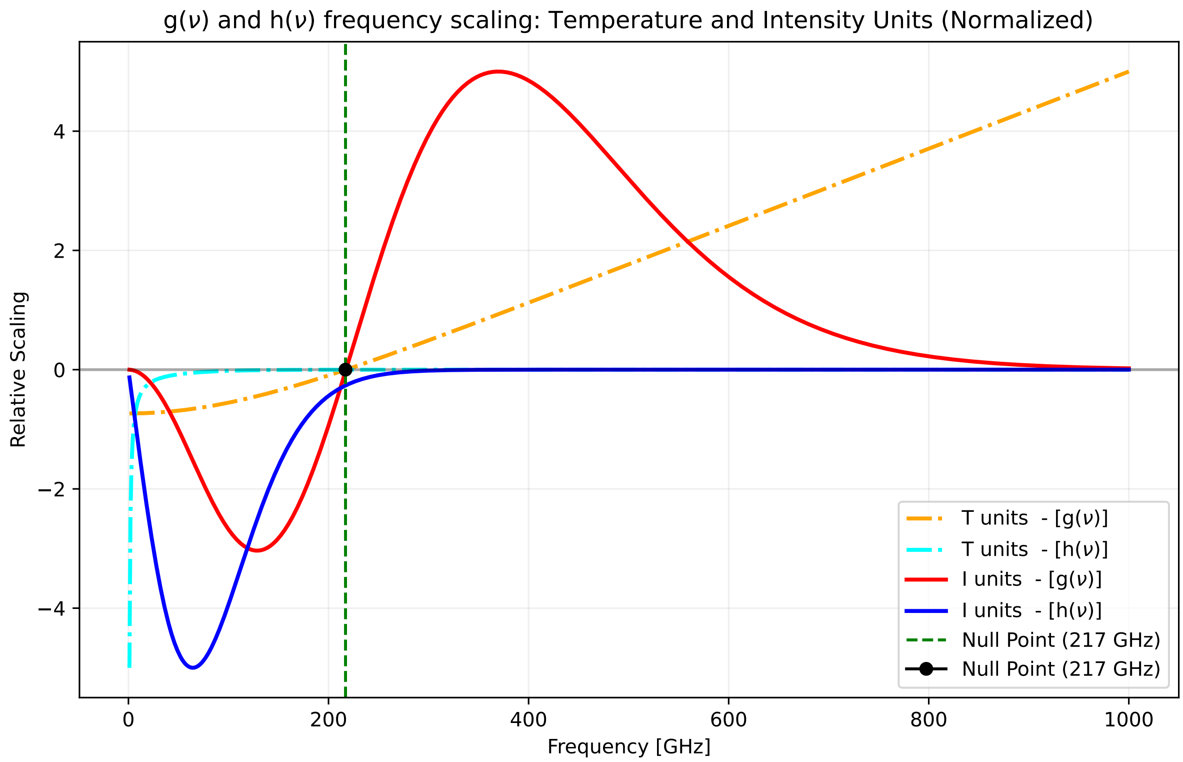

Figure 2.1: Frequency scaling functions for the thermal SZ effect $[g(\nu)]$ and the μ-distorted SZ effect $[h(\nu)]$, as described in Zegeye, Crawford, and Hu, 2024??? sure???. in temperature and intensity units. The null point is around 217 GHz for the thermal SZ effect, while the μ-distorted SZ effect does not have a null point. REDO THE PLOT WITH THE LATEX FORMAT AND CHANGE THE LEGEND MAKINA.

To select a noise sensitivity level for the simulations, I considered a slightly optimistic estimate of the expected sensitivity of future CMB experiments, which is of the order of less than a μK-arcmin. It is frequency independent for simplicity, and it is applied to all frequency channels. After converting this sensitivity level to the appropriate units for the maps, it ends up being even lower, specifically 0.1164 μK-arcmin, which is a very low noise level that allows us to test the performance of the component separation methods in a favorable scenario.

As this thesis intends to explore the ILC and CILC methods for isolating the μ-distorted SZ signal, the simulations do not pretend to be ultra-realistic, but rather to provide a controlled environment to test these methods and understand their limitations and challenges.

The simulations in this work were developed and evaluated across three distinct configurations of frequency channel sets, representing different experimental setups and cosmological baselines:

Configuration	**Frequency Channels (GHz)**	**Total Channels**
Planck	30, 44, 70, 100, 143, 217, 353, 545, 857	9
Zegeye	30, 40, 90, 150, 220, 270	6
PICO	21, 25, 30, 36, 43, 52, 62, 75, 90, 108, 129, 155, 186, 223, 268, 321, 385, 462, 555, 666, 799	21

Table 2.1: Frequency channel configurations used in the component separation pipeline.

The temperature of the CMB is set to $T_{CMB} = 2.7255$ K, and the conversion factor from Compton-y to μK is calculated as $T_{CMB} \times 10^6$.

The main sources of contamination for the μ-distorted SZ signal in these simulations are the thermal SZ effect and the instrumental noise, as there are no foregrounds included in the simulations.

3. Methodology

Methodology

3.1 ILC and CILC

To extract the Cosmic Microwave Background (CMB) from multifrequency maps, the Internal Linear Combination (ILC) method has been widely adopted, although it presents non-negligible local biases Eriksen et al., 2004. To overcome these limitations, advanced modifications such as the constrained ILC (CILC) allow a clean separation of the CMB and the thermal Sunyaev-Zeldovich (tSZ) effect Remazeilles, Delabrouille, and Cardoso, 2011. Furthermore, future missions aim to detect primordial μ-distortions by analyzing the scattering signals of galaxy clusters Zegeye, Crawford, and Hu, 2024.

3.2 Stacking and TT-plot

3.3 Cross- Vs. Auto- Power Spectrum Ratio

4. Results

Results

4.1 ILC and CILC

4.2 Stacking and TT-plot

4.3 Cross- Vs. Auto- Power Spectrum Ratio

5. Conclusions

Conclusions

References

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Zegeye, David, Thomas Crawford, and Wayne Hu (Mar. 2024). "Measuring μ distortions from the thermal Sunyaev-Zeldovich effect". In: 109.6, 063503, p. 063503. DOI: `10.1103/PhysRevD.109.063503`. arXiv: `2307.16853 [astro-ph.CO]`.

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A. Extra plots

I can use this appendix to include any additional images that are relevant to the content of the thesis but were not included in the main chapters due to space constraints.

B. Extra tables

In this appendix, I can include EXTRA numerical results.

c_{sc}^2	c_{sp}^2	v_{Ac}^2	v_{Ap}^2	k_z
12	166	60	1400	0.01

Table B.1: Valors numèrics de velocitats adimensionals realistes i constant d'acoblament del sistema.

C. Code

To find all the code used to obtain the results presented in this thesis, you can visit my GitHub repository (Ribot, 2026). There, you will find all the Python notebooks used for the simulations, calculations, analysis, and plotting of the results.